Design and implementation of user information sharing system using location-based services for social network services*

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Abstract: Internet takes a role as a place for communication between people beyond a space simply for the acquisition of information. Recently, social network service (SNS) reflecting human's basic desire for talking and communicating with others is focused on around the world. And location-based service (LBS) is a service that provides various life conveniences like improving productivity through location information, such as GPS and WiFi. This paper suggests an application combining LBS and SNS based on Android OS. By using smart phone which is personal mobile information equipment, it combines location information with user information and SNS so that the service can be developed. It also maximizes sharing and use of information via twit based on locations of friends. This proposed system is aims for users to show online identity more actively and more conveniently.

Key words: android OS; social network service (SNS); location-based service (LBS); Google maps; Twitter; Open API

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As various services using user's location become possible through development of mobile communication and mobile equipment, the location-based service is being magnified as a new content. The location-based service (LBS)[1] is a service that provides users with diversity based on current location information obtained from networks of mobile carriers or GPS[2] and so on. It is considered as a key factor of ubiquity to provide useful living information, such as weather, stores, amenities and traffic reports, etc. based on where the user is. Meanwhile, as Google decides to provide main services like Google maps, etc. at no cost by connecting to advertisement, 80% of smart phone users are expected to use LBS around 2013. And it is expected that world mobile LBS market will grow rapidly from $111 million in 2008 to $2.2 billion in 2013, which means average 43% growth per year as shown in Fig. 1[3].

Diverse services combining LBS, such as Gowalla[4], FourSquare[5], I'm in[6] and social networking service[7] are unveiled. It increases possibility of access to the information that I need through high timeliness. To reflect these tendencies, this paper maximizes sharing and using of information via twit based on locations of friends. So the system is proposed to help users show online identity more actively and more conveniently.

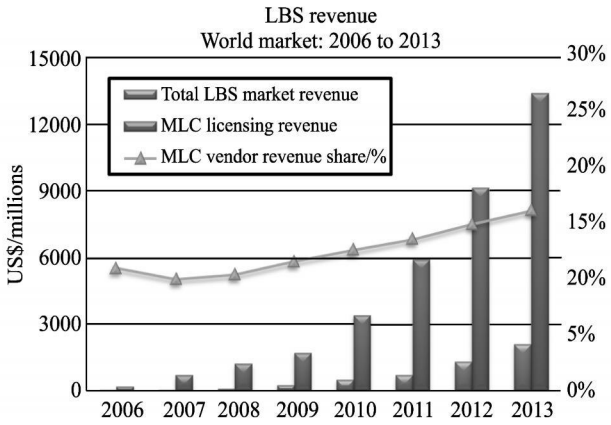

Fig. 1 Market volume of location-based services[3]

1 Related work

1.1 FourSquare

FourSquare service is to share diverse information about user's location and places dropped by between social media members. With introducing the concept of Check-In, transmitted information of more than 1.4 million places and over 1.5 million locations is being recorded. The members are allowed to upload location-based information in the world.

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Distinctively, game-like elements are added, such as giving badges and points according to how many times to visit a certain location. Diverse cases applying FourSquare to marketing are increasing recently.

1.2 OAuth[8]

Authentication is required to take advantage of Open API. Since no standard of authorization existed before using OAuth, ID and password are used that was an existing basic authorization method, which is a vulnerable structure from security point of view. If using OAuth protocol, the website holding user's data can share stored personal information and resources with other sites without exposure of user's personal information. By using OAuth authorization protocol, protection of user's personal information and security can be enhanced. Fig.2 is an authentication process of twitter service. Consumer means a developed application using API and service provider is a provider offering API.

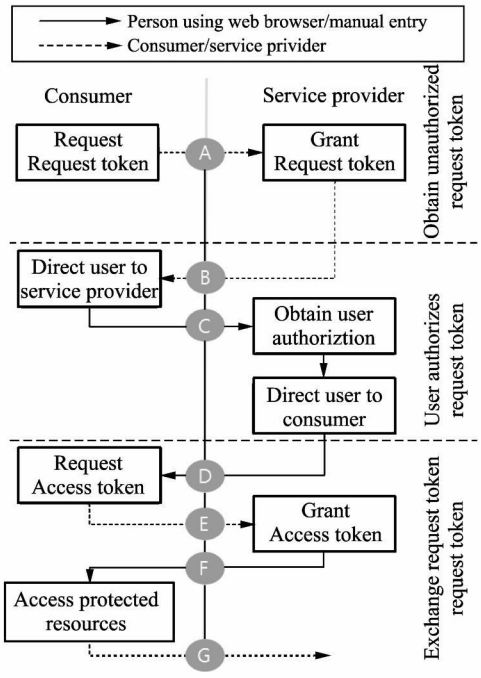

Fig.2 OAuth authentication flow

1.3 Android location package[1,9]

Android gives various functions for supporting location-based service through android. location package as shown in Fig.3.

LocationListener interface within package is a callback function sensing movement and change of state, which contains 4 methods (Table 1). Redefinition is applied for it.

Geocoder class is to provide Geocoding service, which converts address into geographic coordinate. The geographic coordinate enables markers and images to put on Google maps. Authentication is required to take advantage of Open API.

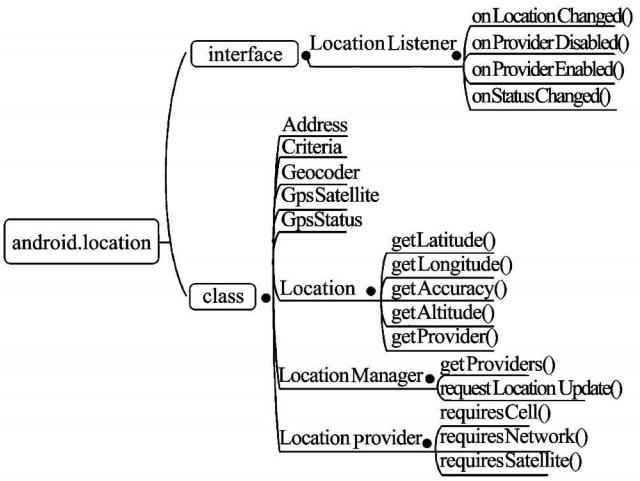

Fig.3 Structure of android location package[1]

Table 1 Items of location listener interface[1]

Method	Description
onLocation-Chagned()	Called when the location changed
onProvider-Disabled()	Called when provider is stopped by user
onProvider-Enabled()	Called when provider is activated by user
onStatus-Changed()	Called when status of provider changed

1.4 Usage of Open API[10]

To utilize Open API, requesting a certain service URL in GET/POST way is required. Ask for the service with assigned Key. Do not forget that Google limits geocode request to 50 000 times per day (Fig.4).

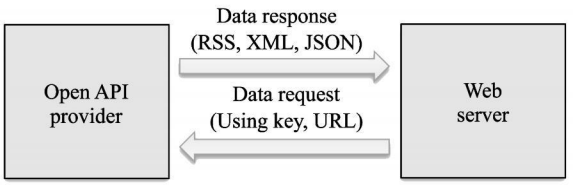

Fig.4 Structure of android location iackage

2 Design and implementation

2.1 System configuration

Our Application was produced based on Android2.1 Eclair. Server is composed of Apache, PHP and MySQL (APM) based on APACHE 2.2.11. Its language used was PHP and database used MySQL5.1.36. The entire system structure is as Fig.5.

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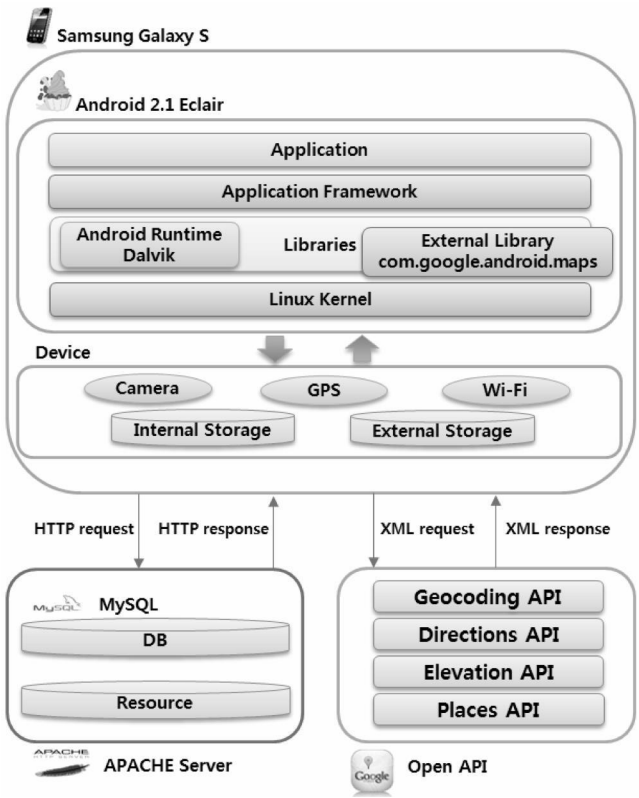

Fig.5 Entire system configuration

Android platform consists of 4 classes. Linux kernel 2.6.29 is in the lowest level. The acquisition of location information is made through device driver of kernel in device. And permission is essential in the highest, which is an application level, to access GPS and wireless internet resource.

Token, user information, current location and state for authorization are stored in the server. Post is used for request to the server and it puts additional information into Header, then transmits it when asked for. There's no limit of length so that it can hand over lots of data. And security is well preserved due to no disclosure of information. data is organized as follows.

1) Account(ID, Password, Name, TwitterID);

2) Authentication(TwitterID, AccessToken, Secret);

3) Friends(ID, Friend);

4) Location((ID, Latitude, Longitude);

5) UserState(ID, State).

The location information senses the location every 10 s. If it senses over 10-m movement, latitude and longitude are supposed to be transmitted to the server. Android map library should be added and applied based on Google map. View is composed through the library, layout and map controller are changed and controlled as well.

2.2 Twitter authentication

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To develop twitter client, Open API offered by Twitter is needed. Java libraries supporting Open API are Twitter4J, JavaTwitter, Jtwitter, Twitter Client and so on. Our system selects and develops Twitter4J that is well-documented and updated steadily.

The application is used after signing up for a membership(Fig.6). Authenticating Twitter service is made through OAuth button. When asking for a RequestToken, running authentication URL, not registering Callback URL additionally, then token issued by Twitter is stored in the server. Token is divided into AccessToken and RequestTokenSecret. AccessToken is a value for consumer accessing protected personal resources not through user's service provider. RequestTokenSecret is a value for authentication that certain access token belongs to the user.

Fig.6 Membership(L)/Twitter authentication(R)

2.3 Main functions

Constitution of application is shown in Fig.7.

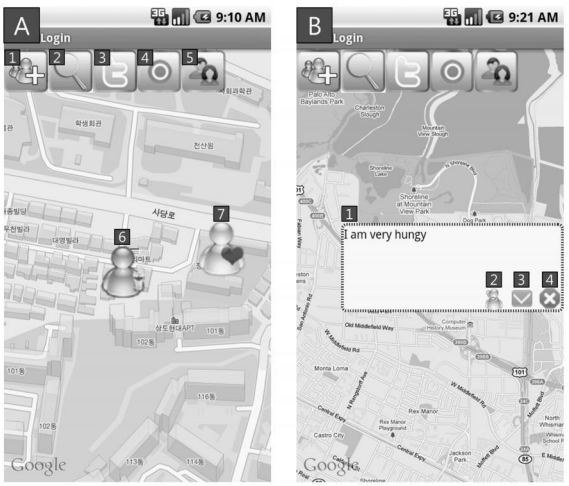

Fig.7 Structure of android location package

Overlay is used to add desired information to View and process touch event. Resource which indi-

cates state information is obtained from the server, which is shown in Overlay. Each function is shown in Table 2.

Table 2 Items of location listener interface

Item(A)	Description
A1	Add users
A2	Search for a particular user's current location
A3	Create Twit (140 character limit)
A4	Search for Particular area using GeoCoding
A5	Set my state
A6	See the latest Twit of friend
A7	My Current Position
Item(B)	**Description**
B1	Twit box
B2	Call a friend
B3	Send SMS
B4	Close

3 Conclusion

LBS is greatly increasing as mobile communication technology is developed and the smart phone, which features mobility and portability, becomes popular. Developing diwerse services becomes convenient through Open API along with opening platform, and network effect gets bigger as problems of information interchange between service providers are improved. Location-based social network service is expected! to become a killer appfirmly in the future. This paper suggests growth potential of location-based social network service, proposing and realizing application and system. People acquire value when they get connected to suitable information in each situation. Development of location-based social network service is required so that it takes advantage of diverse information effectively in mobile equipment.

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